

Classification-Based Authentication Between Fresh Cooking Oil and Bleached Waste Cooking Oil Using an Electronic Nose

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Abstract— Consuming too much leftover cooking oil can cause diabetes, obesity and cancer, among other health problems. Used cooking oil usually has different physical and odor characteristics than fresh cooking oil. However, when purified with bleaching earth and activated carbon, used cooking oil can take on physical characteristics identical to those of fresh cooking oil, making visual and olfactory differentiation difficult. This provides an opportunity for illegal businesses to resell used cooking oil as new, posing a health risk to consumers. Experiments have shown that machine learning algorithms can assist in classifying oil into two categories as bleached waste cooking oil and fresh cooking oil. Among the methods tested, the Random Forest algorithm demonstrated the best performance, achieving an accuracy of 0.9947, precision of 0.9950, recall of 0.9947, and an F1 score of 0.9948. These results indicate that the random forest algorithm is effective in supporting an electronic nose system in identifying and distinguishing bleached waste cooking oil from fresh oil. Thus, this system offers an effective solution for monitoring cooking oil quality, helping to prevent the distribution of potentially harmful waste cooking oil in the market.

Keywords— Classification, electronic nose, waste cooking oil, fresh cooking oil, bleaching oil.

I. INTRODUCTION

The development of palm oil products has a promising future, driven by the expansion of plantations. In Indonesia, land-use changes over 30 years have resulted in the loss of 40 million hectares of forest cover, about 30% of the total forest area [1], [2]. The significant potential of crude palm oil (CPO) and derivatives is fueling this growth, with Indonesia and Malaysia producing 90% of global palm oil exports and accounting for 80% of world production [3]. CPO is extracted from the outer flesh of the palm fruit, while palm kernel oil is extracted from the inner kernel [4]. CPO is rich in palmitic acid (44%), stearic acid (4.5%), oleic acid (39.2%) and linoleic acid (10.1%) [5]. It accounts for 30% of global oil production and is essential for both food and non-food products such as cooking oil, biodiesel, and cosmetics.

Cooking oil, a primary household product derived from CPO, is particularly important to Indonesians and contributes significantly to the national economy and food industry [6], [7], [8]. Indonesia, the second largest producer of palm oil, sets quality standards for CPO through its SNI. The natural bleaching process produces edible oil that meets these standards, characterized by a clear yellow color, 30.664 ppm β -carotene, 0.59% moisture, 0.53% free fatty acids, and a peroxide value of 0.8 mgO₂/100 grams [9], [10].

Despite regulations, fraud in the oil sector is nevertheless common, especially when it comes to selling inferior oil [11]. The bleaching and reselling of recycled cooking oil, which makes it difficult to distinguish from virgin oil due to its visual similarity, is a significant worry. Because frequent heating or recycling greatly raises the peroxide value, speeds up lipid peroxidation, produces harmful reactive oxygen species, and depletes antioxidants in palm oil, consuming such oil offers major long-term health hazards [12], [13]. Additionally, dangerous polycyclic aromatic hydrocarbons (PAHs), which are connected to cancer, are produced during high-temperature reheating. The genotoxic, mutagenic, and carcinogenic qualities of waste cooking oil (WCO) and its vapors result in chromosomal damage and cellular abnormalities in a dose-dependent manner. Excessive WCO consumption has been associated with colon, breast, prostate, and lung cancer [14-16]. Nevertheless, fresh oil and WCO have different aromas and chemical structures that could help distinguish bleached waste oil from fresh oil.

The electronic nose (eNose) uses gas sensor reactions to identify and detect various gases by analyzing multivariate response patterns [17]. eNose is applied in industries like security monitoring, gas leak detection, air quality control, food and beverage testing, then oil quality [18], [19]. Simulating human olfactory senses, eNose can differentiate between the scents of fresh and bleached used cooking oil. However, interpreting its complex data output, which often includes outliers, can be challenging. Machine learning is essential for accurate data interpretation [20], [21].

The novelty of this paper is the use of eNose data classification to discriminate between new cooking oil and bleached used cooking oil using machine learning. Several machine learning methods are used for classification, evaluated, and then the best method is selected. This approach will enable the eNose system to differentiate between bleached used oil and new oil by recognizing the characteristic aromas with the gas sensors in the electronic nose.

II. LITERATUR REVIEW

A. Fresh Cooking Oil and Waste Cooking Oil

Palm oil is made from the mesocarp of the oil palm tree's ripe fruit, which produces thorny clusters of drupes [22]. Crude palm oil (CPO), used for centuries as food and medicine, is rich in palmitic acid, β -carotene, and vitamin E. CPO is often fractionated into palm olein (liquid) and palm stearin (solid) to widen its food applications. Its tocotrienols and β -carotene provide high frying stability [23], [24]. Deep-fat frying, commonly done with refined or fresh oils, retains more nutrients but offers less heat stability [25]. Fried foods are popular for their flavor, texture, and aroma but have high fat content, raising calorie intake and leading to hypercholesterolemia and obesity, which can cause cardiovascular issues. Regular consumption is linked to cancer, heart disease, type 2 diabetes, and high cholesterol [26], [27], [28].

B. Electronic Nose

Partially selective chemical gas sensors are used by electronic nose (eNose) technology to identify and detect a variety of substances in gas mixtures, including volatile organic molecules. eNose is a popular tool for environmental monitoring and the analysis of complex scents in the food sector since it is portable and reasonably priced. By simulating biological olfaction with bionic detecting equipment, the eNose, sometimes referred to as an artificial olfactory system, allows for sensitive and non-destructive odor analysis [29]. However, interactions like compensation, masking, and particularly synergism—where coupled substances give a stronger scent than separate components make it difficult to reliably recognize and classify odors [30].

C. Machine Learning For Classification

Machine learning (ML) is a broad interdisciplinary field that builds on concepts from computer science, statistics, cognitive science, engineering, optimization theory, and other mathematical and scientific domains. One machine learning technique is categorization [31]. Many scientific disciplines have made extensive use of classification systems. This research focus to five method classification.

C.1. Random Forest

The approach combines the results of several decision trees to produce a final prediction for classification using Random Forest. The class with the most votes is chosen as the final output, with each tree in the forest contributing a classification result. Equation 1 describe about random forest basic calculate [32].

$$y = \text{mode}\{f_i(x)\} \quad (1)$$

Where:

y : The final predicted class for the input x .

$\text{mode}\{f_i(x)\}$: The most frequently occurring class among the predictions from all trees.

$\{f_i(x)\}$: The set of predicted classes from each tree i (where $i = 1, 2, \dots, k$)

k : The total number of trees in the forest.

C.2. Decision Tree

Decision tree is a supervised learning method that uses a tree-like structure to represent decisions and their results in classification and regression problems. To produce as homogeneous subsets as possible, the splits are established using metrics such as entropy for classification. Although decision trees are simple to comprehend and analyze, if they grow too intricate or deep without appropriate pruning, they may overfit the training set [33]. Equation 2 is pattern for entropy calculate.

$$\text{Entropy} = -\sum_{i=1}^n p_i \log_2(p_i) \quad (2)$$

Where:

Entropy: The measure of randomness or disorder at a node.

p_i : The proportion of instances belonging to class i at that node.

n : The number of classes.

C.3. Support Vector Machine

Support Vector Machine (SVM) is a supervised machine learning technique that is primarily used for classification problems, while it may also be applied to regression tasks. SVM aims to maximize the margin between data points of various classes, which is the distance between the nearest data points from each class (referred to as support vectors) and the hyperplane [34]. Equation 3 SVM pattern for algorithm uses the following objective function to find the optimal hyperplane:

$$\min \frac{1}{2} \|w\|^2 \quad (3)$$

Subject to the constraint:

$$y_i (w \cdot x_i + b) \geq 1, \forall i$$

Where:

w : The weight vector that defines the orientation of the hyperplane.

b : The bias term that shifts the hyperplane.

x_i : The feature vector of the i th data point.

y_i : The class label of the i th data point (+1 or -1).

$\|w\|$: The Euclidean norm of the weight vector, representing the margin width.

C.4. Naive Bayes

Naive Bayes is a machine learning algorithm based on Bayes' theorem, used for classification tasks. Naive Bayes is particularly effective for high-dimensional data and is used in various applications such as spam filtering, sentiment analysis, and document classification [35]. Equation 4 is basic calculate for naïve bayes calculate.

$$P(C|X) = \frac{P(C|X) \cdot P(C)}{P(X)} \quad (4)$$

Where:

$P(C|X)$: The posterior probability that an instance belongs to class C given features X .

$P(X|C)$: The likelihood, or the probability of features X occurring given class C .

$P(C)$: The prior probability of class C .

$P(X)$: The total probability of features X (a constant for all classes and can be ignored in classification).

C.5. XGBoost (Extreme Gradient Boosting)

XGBoost is a potent and effective open-source machine learning algorithm that is a member of the gradient boosting family. XGBoost iteratively constructs a decision tree ensemble, with each new tree attempting to fix the mistakes caused by the ensemble of earlier trees. Equation 5 is the algorithm minimizes a custom regularized objective function that balances model accuracy and complexity [36].

$$Obj = \sum_{i=1}^n l(y_i, y'_i) + \sum_{k=1}^K \Omega(f_k) \quad (5)$$

Where:

$l(y_i, y'_i)$: A differentiable loss function that measures the difference between the predicted value y_i and the actual value y'_i .

$\Omega(f_k)$: A regularization term to penalize the complexity of the model and prevent overfitting.

Classification algorithms can be evaluated in a variety of ways. To assess various learning algorithms, it is necessary to appropriately understand the study of these metrics and their importance. The frequency with which a model produces accurate predictions throughout the whole dataset under test is known as accuracy. Accuracy offers a comprehensive view of the model's performance by taking into account accurate predictions for both positive and negative classes. Equation 6 illustrates the accuracy calculation process [37].

$$Accuracy = \frac{tp+tn}{tp+fp+tn+fn} \quad (6)$$

Where:

tp (true positive): the number of correct positive predictions.

tn (true negative): the number of correct negative predictions.

fp(false positive): the number of incorrect positive predictions.

fn (false negative): the number of incorrect negative predictions.

Precision gauges how well the model predicts samples that are actually positive. It shows how trustworthy the model's optimistic forecasts are. High precision indicates that the majority of positive predictions are accurate, which is important in circumstances like disease diagnosis when incorrect positive predictions could have serious negative effects. The precision calculation approach is shown in Equation 7 [37].

$$Precision = \frac{tp}{tp+fp} \quad (7)$$

The model's recall gauges its capacity to identify every positive sample in the data. This measure shows how many positive samples the model is able to identify. Equation 8 illustrates how recall is calculated [38].

$$Recall = \frac{tp}{tp+fn} \quad (8)$$

The F1 score is calculated by taking the harmonic mean of recall and precision. This measure is helpful for striking a balance between these two metrics, particularly when recall and precision have highly disparate values. Equation 9 illustrates how the F1 score is determined [38].

$$F1\ Score = \frac{2 \cdot precision \cdot recall}{precision+recall} \quad (9)$$

III. PROPOSED METHOD

This study will discuss the data collection process and the classification of differences between fresh oil and bleached waste cooking oil. Fig. 1 illustrates the research workflow.

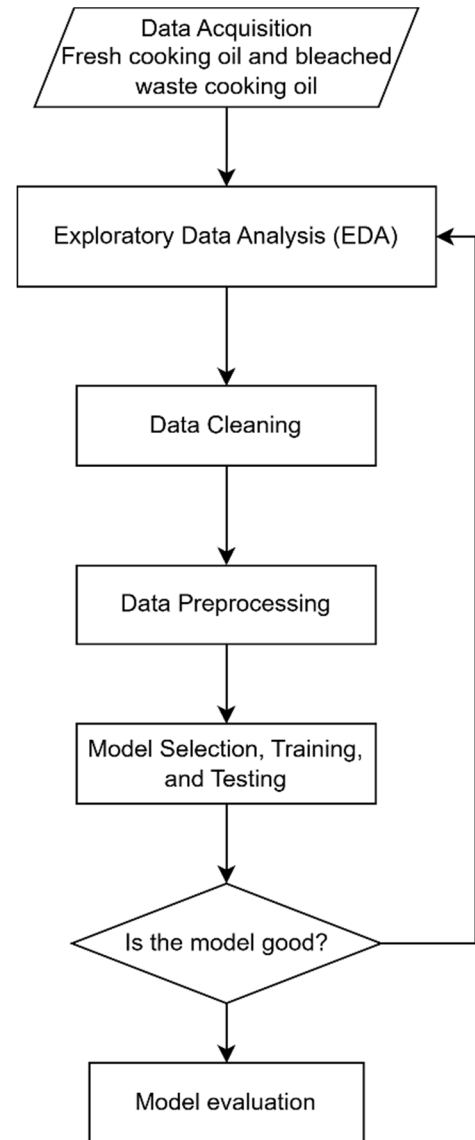


Fig. 1. Proposed Method

A. Data Acquisition

Data acquisition was conducted using an electronic nose with both fresh cooking oil and waste cooking oil that had undergone the bleaching process. Below is an explanation of the data acquisition process:

A.1. Bleaching Waste Cooking Oil Production

Waste cooking oil (WCO) undergoes refining to enhance quality and remove impurities. The process starts with a 60-hour thermal treatment, involving 15-minute heating cycles followed by 45-minute cooling periods, causing physical changes in color and aroma. The bleaching phase follows standard procedures using activated carbon and bleaching earth. WCO is heated for 5 minutes, mixed with activated carbon, cooled, and filtered. It is then reheated, mixed with bleaching earth, and left for 24 hours to adsorb remaining contaminants before a final filtration, resulting in a clearer product.

A.2. Electronic Nose for Data Acquisition

The electronic nose, which is made up of eight sensors that are sensitive to the gases created, works by simulating the human sense of smell. In order to perform the data acquisition system, the baseline reading of the ambient air is first taken, followed by a switch to the target object for detection and a subsequent return to record the free air data. The next step involves processing the acquired readings. Fig. 2 show working flow electronic nose.

B. Data Sheet and Exploratory Data Analysis

The data set for this study consists of 30 samples: 15 used cooking oil samples and 15 fresh cooking oil samples. Each sample contains 150 data points recorded using an electronic nose designed to capture distinct olfactory profiles. An initial exploratory data analysis (EDA) is performed to identify patterns and address any irregularities. Outliers are managed using Z-score analysis to detect significant deviations, and a moving average algorithm is applied to smooth the data and reduce noise.

After data cleaning, preprocessing steps such as normalization, scaling, and transformation are performed to prepare the data for machine learning model training. This ensures improved model performance and reliability. The refined dataset is then used for model selection to identify the best algorithms for accurately distinguishing between waste and fresh cooking oil samples.

C. Model Selection

Machine learning algorithms will be used in this project to categorize data using a range of classification techniques that are most appropriate for the dataset. Random Forest, Decision Tree, Support Vector Machine (SVM), Naive Bayes, and XGBoost are the techniques used for this comparison. These techniques were picked because of their special advantages and versatility in handling various kinds of data, which makes them perfect for assessing the dataset from various analytical angles.

Accuracy, F1 score, precision, and recall are the main metrics that will be used to evaluate each model's performance in order to guarantee a comprehensive analysis. These measures offer a fair assessment of each model's

advantages, demonstrating both its ability to forecast outcomes and its ability to control for class imbalances in the data. By examining and contrasting these outcomes, we want to determine which model is best suited for this particular categorization task and optimize it to provide the highest prediction performance.

IV. RESULT AND DISCUSSION

A. Data Visualization

The electronic nose generates digital data that reflects the sensor responses after each sensor has been preconfigured or calibrated to identify specific characteristics of the substances being tested. Measurements of free air are used to represent baseline conditions in the first and third segments of the instrument's data collection, while measurements of air that has come into contact with the sample are recorded in the second segment. Data from a cooking oil sample is shown graphically in Figure 3, which illustrates how molecular interactions between the oil and sensor components affect sensor responses. An example of data acquisition is shown in Table I, which also provides information on the volatile profile or chemical composition of the frying oil sample. It is expected that this presentation will enhance the understanding of the sample composition.

B. Waste Cooking Oil After Bleaching

Waste cooking oil was purified by bleaching with fuller's earth and activated carbon. The purpose of this purification process was to study in detail how the physical properties and the odour of the oil changed after these procedures. The three different types of oil studied have different characteristics, as shown in Figure 4. The fresh cooking oil (2) looks transparent and smells fresher and more distinctive than the original waste cooking oil (3), which is comparatively black in color and has a strong distinctive perfume. The color of the used cooking oil became clearer and more like fresh cooking oil after the bleaching process (1), and its characteristic odor also changed. This bleaching process produces significant visual and aromatic results in used cooking oil, bringing its physical characteristics and aroma closer to those of fresh cooking oil.

C. Classification Result

During the training and evaluation, various classification models were applied to sensor data from two groups (used oil and new oil), combined into a single dataset. The dataset was split into features (X) and target labels (y). The models tested included Random Forest, Decision Tree, SVM, Naive Bayes, and XGBoost, with hyperparameter tuning done using RandomizedSearchCV and 5-fold cross-validation based on accuracy.

The evaluation results in table II indicated random forest had the best performance with an accuracy of 0.9947, precision of 0.9950, recall of 0.9947, and F1 Score of 0.9948. SVM and XGBoost also performed well, with accuracy above 0.96, while Naive Bayes and Decision Tree showed slightly lower results. Metrics used for evaluation included accuracy, precision, recall, and F1 Score.

The SVM model used a linear kernel, $C=10$, and $\gamma=0.1$. The random forest model had 100 estimators, with a minimum of two samples per internal node and per leaf, and no max depth. The decision tree had a max depth of

30, four samples per leaf, and a minimum of 10 samples per split. XGBoost was configured with a subsample ratio of 1.0, 100 estimators, a max depth of 10, and a learning rate of 0.2. The naive bayes model assumed a gaussian distribution for continuous features.

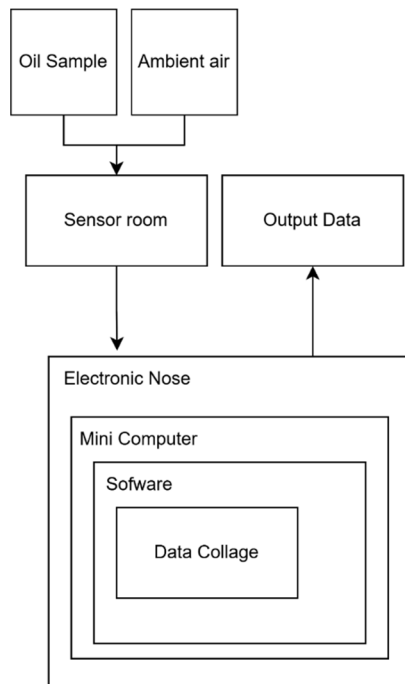


Fig. 2. Working Flow Electronic Nose

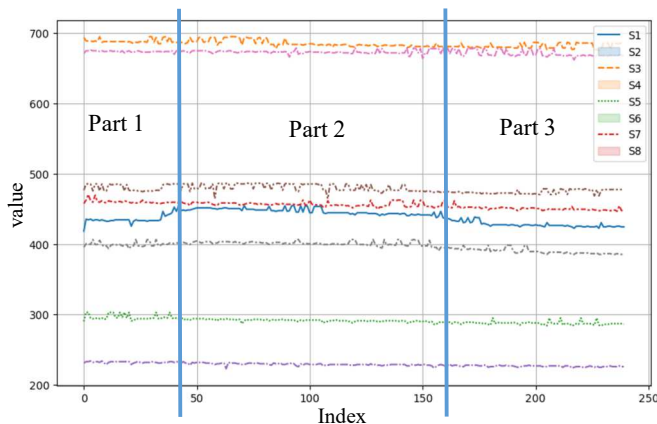


Fig. 3. Data Visualization



Fig. 4. Waste Cooking Oil after Bleaching

TABLE I. DATA SAMPLE FROM ELECTRONIC NOSE

S1	S2	S3	S4	S5	S6	S7	S8
382	597	241	376	215	559	648	330
402	600	241	374	209	555	652	340

403	602	244	378	211	555	653	351
404	603	245	377	211	555	662	344
403	602	243	376	210	554	653	342
404	603	245	377	212	554	653	342
405	604	244	377	212	555	661	343
405	604	244	378	212	556	653	344
405	604	243	376	211	554	652	343

TABLE II. RESULTS OF CLASSIFICATION EVALUATION

Model	Accuracy	Precision	Recall	F1 Score	Best Parameters
Random Forest	0.9947	0.9950	0.9947	0.9948	n_estimators=100, min_samples_split=2, min_samples_leaf=2, max_depth=None
Decision Tree	0.9464	0.9561	0.9464	0.9460	min_samples_split=10, min_samples_leaf=4, max_depth=30
SVM	0.9942	0.9946	0.9942	0.9943	kernel='linear', gamma=0.1, C=10
Naive Bayes	0.9672	0.9748	0.9672	0.9669	parameters GaussianNB
XGBoost	0.9692	0.9760	0.9692	0.9680	subsample=1.0, n_estimators=100, max_depth=10, learning_rate=0.2

V. CONCLUSION

Differentiating between fresh and bleached used cooking oil is a significant barrier to quality control in the food industry. Because they look alike, traditional detection techniques often fail, and reusing bleached oil can be hazardous to health. Creative solutions are needed to ensure public safety.

Effective separation of fresh from bleached waste oil was demonstrated in an experimental investigation that combined machine learning models Random Forest, SVM, and XGBoost with an electronic nose. With a 99.48% F1 score, 99.47% accuracy, 99.50% precision, and 99.47% recall, the Random Forest model outperformed the other models. This shows how machine learning can improve quality control.

Future advancements could focus on improving the electronic nose's sensitivity and expanding datasets to include diverse oil samples, boosting model robustness. Real-time detection system integration into production lines could further streamline quality control, ensuring ongoing safety improvements in food processing.

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